Is TunisianMMLU Actually Tunisian?

A Native-Speaker Audit of a Machine-Translated Dialect Benchmark

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The dominant defect is not translation noise but a different dialect.

A scan flags 63.4% of the 21,500 items, and two native speakers mark between 62.0% and 86.0% of a 400-item sample defective. Corrected text raises every above-chance model by 2.3 to 4.9 points, and one gain survives multiple-test correction.

The benchmark

- **TunisianMMLU (Ghezaiel and Lorré, 2025):** 21,500 multiple-choice items in 44 subject configurations, translated from MMLU, ArabicMMLU and DarijaMMLU with NLLB-200-3.3B, followed by CODA normalization; no released item is marked as DarijaMMLU-derived.
- **Its own documentation:** annotations_creators: machine-generated, with no native validation stage listed.
- **To our knowledge** the only MMLU-style benchmark for Tunisian Arabic (Derja); released by the developers of the Labess Tunisian LLM to evaluate their own model in-dialect.

Does it measure Tunisian language competence, or translation artifacts?

Two checks

Corpus scan. All 21,500 items are scanned for mechanically detectable Moroccan markers and pipeline artifacts, with native-validated patterns; Tunisian and MSA words sharing the same initial letters are excluded by list.

Native-speaker audit

- 400 items stratified over all 44 subjects (a floor of four per subject, seed 20260805).
- Two native Tunisian speakers, blind and independent; annotator B was unaware of the hypotheses.
- Six binary labels per item: semantic fidelity, answer preservation, Tunisian authenticity, MSA/Moroccan interference, orthographic naturalness, cultural validity.
- Verdict OK, FIXABLE (repairable by a native speaker in under 90 seconds, a guideline rather than a timed measurement) or DISCARD (unrepairable without retranslation); origin TRANSLATION, SOURCE or UNCLEAR.
- Disagreements adjudicated by annotator A, an author.

A FIXABLE item (driving test)

Original

إلا ماكانوش لوحات كيحددو السرعة، السرعة القصوى المسموح بيها داخل المناطق المأهولة كتكون:

Moroccan ka- present prefix (كيحددو, كتكون) and ماكانوش existential negation.

Corrected

إذا ما فماش علامات تحدّد في السرعة، السرعة القصوى المسموح بيها داخل المناطق العمرانية هي:

Tunisian existential ما فماش and bare imperfect تحدّد.

"If there are no signs specifying the speed limit, the maximum speed allowed inside built-up areas is:"

Corpus scan: 63.4% of items flagged

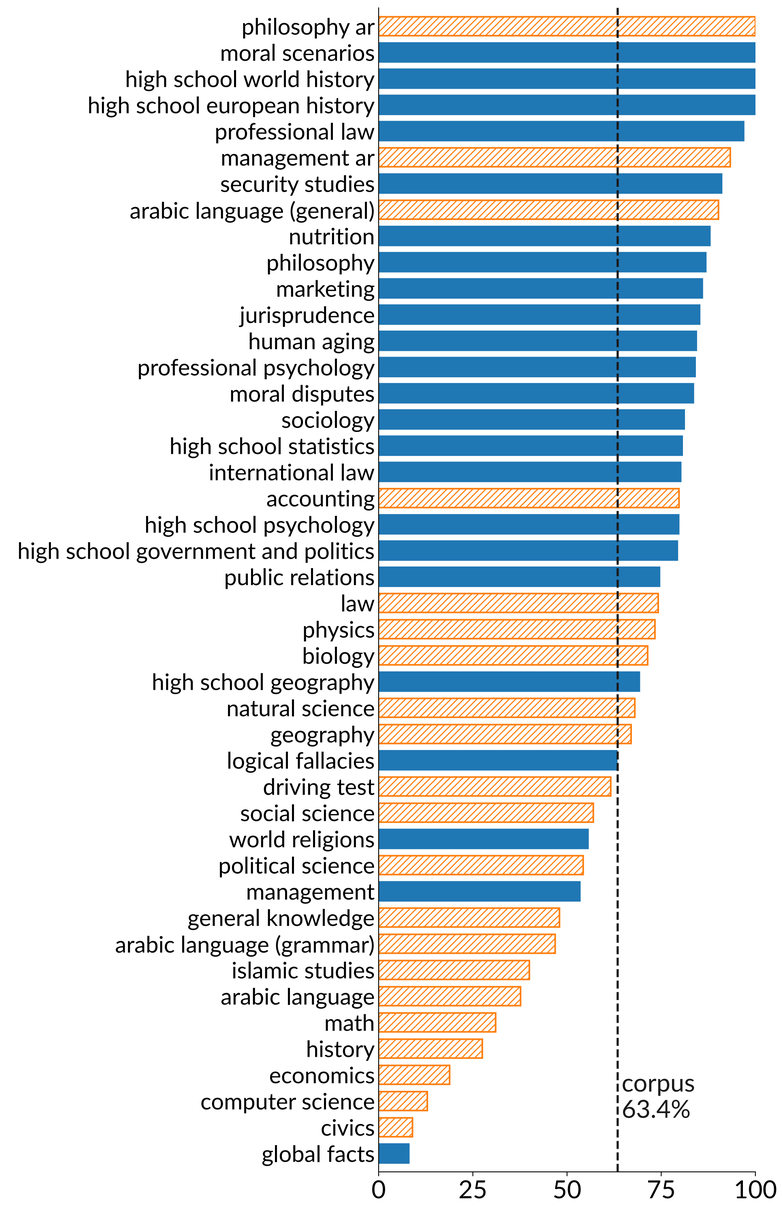
55.1% of items carry the Moroccan present-tense prefix. Tunisian Derja marks the present with the bare imperfect and has no such prefix.

Present-tense prefix	كت-/كي-
Moroccan function words	ديال، غادي، كاين، بغا، واخا، يدير
Tunisian counterparts	فما، متاع، باش
Single-lam spelling of اللي	الي
Pipeline artifacts	فيي and a literal \n

Several lexical markers are shared with Algerian varieties, and غادي is also a Tunisian locative adverb; the attribution to Moroccan rests on the prefix.

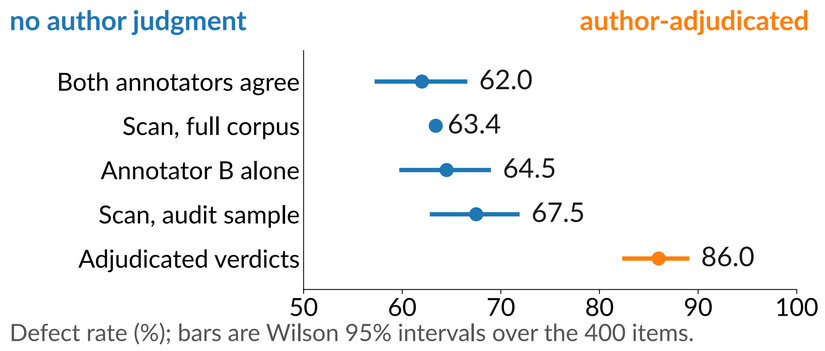
Items flagged by the scan in each of the 44 subject configurations (%)

MMLU-derived: 86.3% ArabicMMLU-derived: 51.6%



- Moroccan morphology on MMLU-derived items cannot be inherited from an English source: unless items are mislabelled by provenance, which we cannot rule out, the pipeline introduced it.
- Eight subject configurations are at or above 90%, including moral_scenarios (895 of 895 items).
- Against the adjudicated verdicts the scan flags 5 of the 56 OK items (98.1% precision, 77.0% recall). It under-counts: absence of a string match does not indicate a clean item.

Native-speaker audit: 62.0% to 86.0% defective

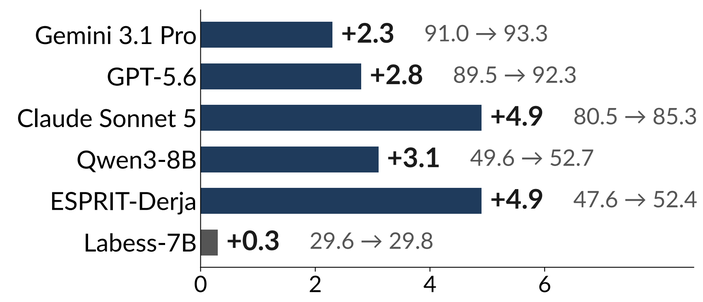


Defect rate (%); bars are Wilson 95% intervals over the 400 items.

- **After adjudication,** 344 of 400 items (86.0%) are defective: 333 FIXABLE, 11 DISCARD, 56 OK.
- **Conservative bound:** both annotators independently agree on 248 items (62.0%); annotator B alone, who is not an author, marks 258 (64.5%).
- **Origin:** 330 of the 344 defects (95.9%) are TRANSLATION-origin, 7 (2.0%) SOURCE-origin; an upper bound on the translation share, since annotators did not consult the sources.
- **Agreement:** raw 73.2%, κ = 0.32. The most objective label, MSA/Moroccan interference, reaches κ = 0.55; the labels that ask whether a Tunisian would produce the text sit lowest, since Derja has no standard orthography.
- Most defective items remain answerable; the benchmark fails on its stated construct rather than as a set of questions.

Corrections help every above-chance model

Gain on corrected text (accuracy points, same 389 items; Δ from item counts before rounding)



- **All five above-chance models** improve on corrected text, by 2.3 to 4.9 points. Labess-7B, the model the benchmark was released to evaluate, is at chance in both conditions.
- **No model changes rank,** but the gaps change: Qwen3-8B's lead over its Derja fine-tune ESPRIT-Derja falls from 2.1 points to 0.3, and ESPRIT-Derja gains more on ArabicMMLU-derived items (+7.7) than on MMLU-derived ones (+1.2).
- **McNemar tests:** only the Claude Sonnet 5 gain stays significant under Holm correction, so we treat the individual gains as suggestive and the narrowing of the gaps as descriptive.
- **Caveat:** corrected text may sit closer to MSA, so part of each gain may be general denoising rather than dialect repair; the absence of any gain for Labess argues against a pure-fluency account.
- **Caveat:** the evaluated corrected file carried a formatting error on 173 of the 333 FIXABLE items; gains on those and on the 160 correctly formatted items differ by less than three points for every model.

Conclusions: four practices for machine-translated benchmarks

Native validation before release. A two-annotator audit of a 400-item sample found between 62.0% and 86.0% of it defective.

A marker scan as a release-time check. A validated string scan covered the full corpus at 98.1% precision and would have caught the dominant defect.

Error-origin labeling, so that translation-introduced damage is not conflated with inherited source errors.

Model accuracy by verdict band. All six models order OK > FIXABLE > DISCARD on the original text, a convergence check available to any audited benchmark; with 11 DISCARD items, that column is indicative rather than precise.

A dialect-adapted model evaluated on text written in the wrong dialect is measured least well on exactly the capability it was adapted for.

In this paper the contrast rests on a single base-model pair, Qwen3-8B and its Derja fine-tune, and we read it as consistent with suppression of the adapted model's advantage, not as a causal estimate.

Released: TunisianMMLU-Verified

Both annotators' labels, verdicts and origins; corrected Derja text for the 333 FIXABLE items; severity flags for all 21,500 items; the guideline; all evaluation code. Keyed to the original item identifiers, so corrections and flags can be merged upstream.

code and annotations

data

Corrected text CC BY-NC-SA 4.0 (inherited from TunisianMMLU); labels and guideline CC BY 4.0; code MIT.

Limitations. Adjudication was performed by one annotator who is an author, so the agreed floor (62.0%) and the non-author rate (64.5%) are reported alongside every adjudicated figure. Open-weight models run under 4-bit quantization; all model results come from a single run with one prompt template; provenance and subject are confounded in the benchmark. Corrections were written by annotator A and not independently verified; the dialect-adaptation contrast rests on a single base-model pair.

Links. github.com/jjlalli/tunisianmmlu-audit and huggingface.co/datasets/fatmajlali/TunisianMMLU-Verified. Reference. Ghezaiel and Lorré (2025). TunisianMMLU. Hugging Face dataset, huggingface.co/datasets/linagora/TunisianMMLU (LINAGORA; accessed August 2026).

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